Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/10/2005
PI Name:	Chan, Moses H.W. Ph.D.		
Project Title:	Critical Casimir Forces		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	16802	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	04/01/2001	**End Date:**	02/28/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG 3 2891		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		
Task Description:	To task objective is being accomplished (1) by constructing new sample cell with silicon wafers as electrodes to measure the thinning of helium film as the cell is cooled through the superfluid transition (2)by constructing a Casimir force apparatus consist of a flexible diaphragm separated at a distance of a micorn from a fixed electrode, allowing the measurement of the Casimir force directly (3) testing the device at cryogenic temperatures. The objective is to measure fluctuation-induced forces in the vicinity of critical points of three different universality class: the liquid gas critical point, the lambda point in He-4 and the He-3-He-4 Mixture Tricritical point. We want to make the measuremen on a film that is adsorbed on a well surface such as silicon thatis atomically flat. We have made measurement on the change in the chemical potential as the system is brought through the critical point, we want in the seond experiment to measure the force directly. One of the most beautiful aspects of physics is how pheonomena in widely different systems are described by the same mathematical formulation. Directly analogous to the Casimir force in electromagnetism, the critical Casimir		

	force is predicted to be a universal feature of all critical points, containing important information about finite size effects and boundary conditions. The result of this work will provide the first measurement of the critical Casimir effect near a liquid-gas critical point and it will allow us to examine the effect of varying boundary condition and different universality class on the properties of fluids near their critical points.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We have succeeded in assembling a sample cell with silicon wafers as the electrodes. In contrast to the version with copper electrodes, the silicon electrodes are atomically flat so that there will not be extra surface area complicating the interpretation of the data. We have cooled down the cell to cryogenic temperature once and the cell survived the test. In the Casimir force apparatus, we have constructed a pair of electrodes that consist of a flexible kapton diaphragm of 2 micron in thickness and 2.5 cm in diameter that is separated from a fixed spherical electrode. By means of a piezo-positioning system (attocube systems) we are able to maintain a gap of less than a micron. Preliminary testing indicates this system should have the sensitivity to measure directly the Casmir force near the lambda point. We have attached the experimental cells to two separate cryostats and we will begin the process of introducing helium into the sample cells to gather data.
Bibliography Type:	Description: (Last Updated: 09/11/2009)
Articles in Peer-reviewed Journals	Garcia R, Scheidemantel S, Knorr K, Chan MH. "Critical adsorpton in a well-defined geometry." Physical Review E 2003 Nov;68(5 Pt 2):056111. PubMed PMID: 14682850 , Nov-2004